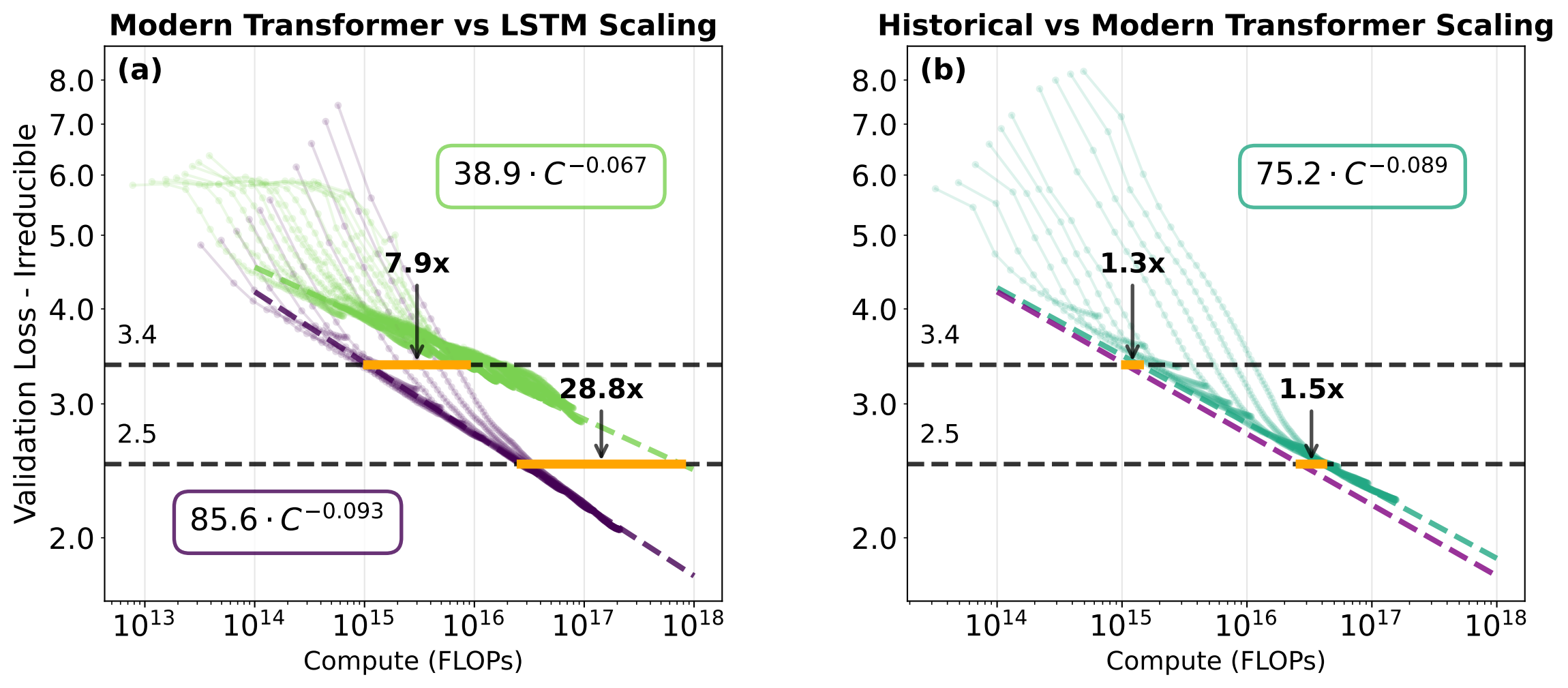

Modern Transformer vs LSTM Scaling
(a)
Validation Loss - Irreducible
8.0
7.0
6.0
5.0
4.0
3.4
3.0
2.5
2.0
$38.9 \cdot C^{-0.067}$
$85.6 \cdot C^{-0.093}$
7.9x
28.8x
10^{13}
10^{14}
10^{15}
10^{16}
10^{17}
10^{18}
Compute (FLOPs)
Historical vs Modern Transformer Scaling
(b)
$75.2 \cdot C^{-0.089}$
1.3x
1.5x